



Exploring the impact of climate variability during the Last Glacial Maximum on the pattern of human occupation of Iberia



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ARTICLE INFO

Article history:

Received 26 November 2012

Accepted 13 June 2014

Available online 15 July 2014

Keywords:

Western Europe

Palaeoclimate

Glacial refugium

Variability selection

Population distribution

Climate modeling

Spatial analysis

ABSTRACT

The Last Glacial Maximum (LGM) was a global climate event, which had significant repercussions for the spatial distribution and demographic history of prehistoric populations. In Eurasia, the LGM coincides with a potential bottleneck for modern humans and may mark the divergence date for Asian and European populations (Keinan et al., 2007). In this research, the impact of climate variability on human populations in the Iberian Peninsula during the Last Glacial Maximum (LGM) is examined with the aid of downscaled high-resolution (16 × 16 km) numerical climate experiments. Human sensitivity to short time-scale (inter-annual) climate variability during this key time period, which follows the initial modern human colonisation of Eurasia and the extinction of the Neanderthals, is tested using the spatial distribution of archaeological sites. Results indicate that anatomically modern human populations responded to small-scale spatial patterning in climate variability, specifically inter-annual variability in precipitation levels as measured by the standard precipitation index. Climate variability at less than millennial scale, therefore, is shown to be an important component of ecological risk, one that played a role in regulating the spatial behaviour of prehistoric human populations and consequently affected their social networks.

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Introduction

Global climate change is a subject of pressing concern as we recognise: a) that it has the potential to fundamentally alter biological systems, including our own, in ways that we do not perfectly understand; and b) that we are unable to accurately predict the timing and severity of extreme events associated with it. Assessing the impact of climate change on a heavily populated planet, therefore, is still a distant goal (cf. the IPCC special report on extremes¹). Climate modeling techniques, originally developed to understand recently observed climate variation and to predict future climate evolution, can also be used to simulate past climates. In fact, palaeoclimate simulations are a necessary step in evaluating the ability of climate simulations to accurately represent situations

that are different from the present state (Braconnot et al., 2012) which is one of the reasons the Palaeoclimate Modelling Inter-comparison Project (PMIP), now in its third phase, was initiated (Otto-Bliesner et al., 2009). The inherent value of the palaeoclimate record for climate modellers is the opportunity it offers for testing models against actual occurrences of abrupt change and investigate model sensitivity to different conditions (Valdes, 2011).

The archaeological record offers a unique opportunity to study climate change and the potential range of human responses to it. During the course of their fieldwork, archaeologists collect palaeoclimate data (e.g., pollen samples, malacological and faunal remains) useful for testing climate model outputs. Collaborations between climate scientists and archaeologists, therefore, have the potential to advance research agendas within both fields of enquiry while contributing to a better understanding of how climate change and climate variability may have influenced key events in human prehistory, such as dispersals into (and within) Europe (Mellars, 1998; van Andel, 2003; Gamble et al., 2004; Agustí et al., 2009;

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¹ Source: http://www.ipcc-wg2.gov/SREX/images/uploads/SREX-All_FINAL.pdf.

Hoffecker, 2009; Verpoorte, 2009; Dennell et al., 2011; Müller et al., 2011; Abbate and Sagri, 2012; MacDonald et al., 2012; Schmidt et al., 2012) and the extinction of the Neanderthals (d'Errico and Sánchez Goñi, 2003; Sepulchre et al., 2007; Golovanova et al., 2010; Bradtmöller et al., 2012; Lowe et al., 2012).

According to the Variability Selection Hypothesis (Potts, 1996, 1998) the hominin lineage has evolved “intricate, canalized adaptive systems” (Potts, 1998:89) that confer widespread plasticity and strong sensitivity to environmental input as a response to long-term environmental change. A key aspect of environmental change, according to this hypothesis, is an increase in climate variability over the past few million years. While the impact of climate variability on human populations has been investigated on a large (millennial) scale, e.g., for the last Glacial in Iberia (Davies and Gollop, 2003; van Andel and Davies 2003; Gamble et al., 2004; Tzedakis et al., 2007), it has yet to be investigated on a finer spatial and temporal scale. Thus, the limits of hominin sensitivity to environmental inputs and the amplitude of climate change required to stimulate an adaptive response are not yet established. Here, we develop a new methodology that enables us to capture fine resolution and high frequency climate variability and test its impact on human systems.

We specifically designed this research to test the sensitivity of human systems to climate variability at different spatial and temporal scales. By climate variability we mean “variations in the mean state, the standard deviations and the occurrence of extremes of the climate, on all temporal and spatial scales beyond that of individual weather events.”² From a climate modeling perspective, we test a new statistical downscaling method that allows us to obtain climate model results at much finer spatial resolution than previously possible. This provides us with detailed palaeo-environmental models with which to critically assess the role of climate-induced environmental change in conditioning human population dynamics. The results of our research will help establish how ecological variability affected the distribution of modern human populations in the Iberian Peninsula during the Last Glacial Maximum (LGM) and suggest what impact this may have had on their social organization.

The target region is a well-defined biogeographical entity (Gamble, 1986; Fernandez et al., 2004) for which a well-documented series of archaeological sites dated to the LGM exists. Several factors make the Iberian Peninsula an interesting region in which to investigate human–environment interactions, including its varied climate, attributable to its geographical position between the Atlantic Ocean and the Mediterranean (O'Regan, 2008) and its probable role as a population reservoir, or refugium, for northern European populations during cold climate phases of the last Glacial period (Straus et al., 2000; d'Errico and Sánchez Goñi, 2003; Gamble et al., 2004; Verpoorte, 2009). The timeframe for this phase of our research, the LGM, is the period of maximum global ice volume which corresponds to the interval between 19 and 23 ka cal. B.P. (thousands of years calibrated before present) (Mix et al., 2001). Archaeologically, this timeframe is significant as it marks a period of probable range contractions for human populations throughout northern Eurasia as well as the appearance of a modern human European morphotype (Churchill et al., 2000).

The archaeological record suggests that human occupation of the Iberian Peninsula during the LGM was spatially discontinuous (Fig. 1). This pattern is usually attributed to the avoidance of high levels of ecological risk associated with extreme climate conditions; cold temperatures and/or dry conditions are frequently

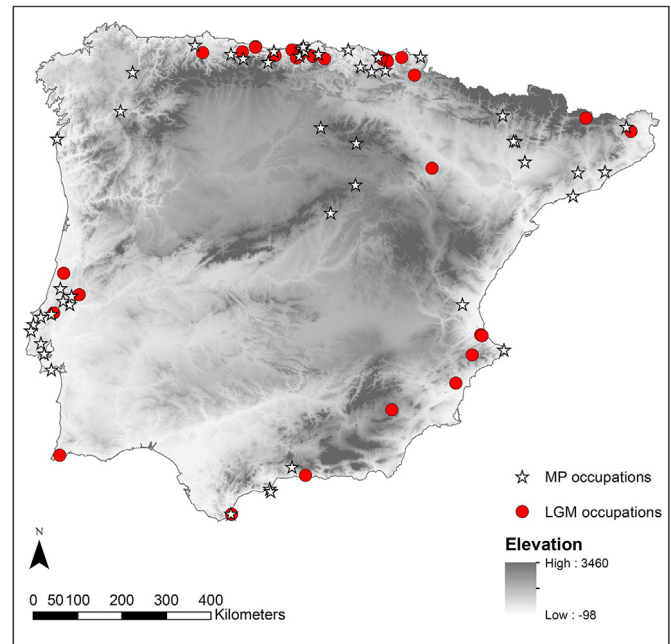


Figure 1. The distribution of archaeological sites used in this research. Sites dated to the LGM (circles) and Middle Palaeolithic (stars).

invoked as variables affecting the distribution of human populations (Cacho et al., 2010; Jennings et al., 2011; Bradtmöller et al., 2012; Schmidt et al., 2012). It has been demonstrated (Davies and Gollop, 2003), however, that average temperature, wind-chill and snow tolerance are not significant predictors of archaeological site location across Western Europe during cold or warm phases of the last Glacial. The potential impact of climate on human spatial behaviour in the Iberian Peninsula during the LGM, therefore, needs to be rigorously tested at different scales of analysis. In addition to testing climate variables such as temperature and precipitation at a finer spatial and temporal scale than possible when Davies and Gollop (2003) performed their analysis, we investigate the role of climate variability, a factor contributing to ecological risk, as a potential factor in controlling the distribution of human populations.

Materials and methods

We generated a climate simulation for the LGM covering the Iberian Peninsula with a spatial resolution of 16×16 km. After testing the results of this numerical climate experiment against environmental proxy data, we quantified climate variability over the target area. Finally, we conducted a spatial statistical analysis of the presence of archaeological sites using logistic regression and model selection.

The archaeological sample

A database of geo-referenced archaeological sites on the Iberian Peninsula (Fig. 1; Table 1) with radiocarbon dates falling within the timeframe of interest was initially drawn from the PACEA database (d'Errico et al., 2011), cross-referenced with the Radiocarbon Palaeolithic Europe database (v14).³ Dates are calibrated using the CalPal-2007-HULU calibration curve (Weninger and Joris, 2008). We verified the stratigraphic provenance of the dating samples and

² Source: <http://www.ipcc.ch/ipcreports/tar/wg1/518.htm>.

³ <http://ees.kuleuven.be/geography/projects/14c-palaeolithic/download/>.

Table 1

The list of georeferenced archaeological sites, dated to the LGM, used in the statistical analysis.

Site name	Longitude	Latitude	Level	Code	Age	Error	Cal BP	Error	Cultural attribution (sample)	References
Abauntz	1.633300	43.001500	E	Land-1965	15,800	350	19,050	390	Magdalenian	Evin et al., 1983; Mazo and Utrilla, 1995
Aitzbitarte IV	1.895600	43.262800	IV	Grn-5993	17,950	100	21,470	90	Solutrean	Straus, 1990; Soto Barreiro, 2001, 2003
Altamira	4.119000	43.378000	III	GifA-90045	18,540	320	22,220	440	Solutrean	Soto-Barreiro, 2001, 2003
			art	GifA-96061	16,480	210	19,770	310	(black mark)	Moure et al., 1997
Amalda	2.204700	43.234200	IV	I-11355	17,580	440	21,130	530	Solutrean	Mariezkurrena, 1990; Straus, 1990
			IV	I-11428	16,200	380	19,470	470	Solutrean	Mariezkurrena, 1990; Straus, 1990
			IV	I-11435	16,090	240	19,290	320	Solutrean	Mariezkurrena, 1990; Straus, 1990
			V	I-11372	17,880	390	21,550	550	Solutrean	Mariezkurrena, 1990; Straus, 1990
Ambrosio	2.099100	37.822200	II	Gif-7276	16,500	280	19,770	380	Solutrean	Ripoll López et al., 1994; Villaverde et al., 1998
			II.2	Gif-A-II.2	19,170	190	22,966	316	Upper Solutrean	Aura et al., 2012
			II.g	GifA 9883	19,250	70	23,017	272	Upper Solutrean	Aura et al., 2012
Arbreda	2.746800	42.161800	3.03–3.15	Gif-6418	17,320	290	20,810	330	Solutrean	Delibrias et al., 1987; Soler and Maroto, 1987; Villaverde et al., 1998
			3.25–3.50	Gif-6419	17,750	290	21,290	360	Solutrean	Delibrias et al., 1987; Villaverde et al., 1998
Beneito	0.466700	38.778000	B2	Land-3593	16,560	480	19,840	590	Solutrean	Villaverde et al., 1998; Aura et al., 2012
Buraca Grande	8.733300	39.983300	9a	Gif-9502	17,850	200	21,380	230	Solutrean	Aubry et al., 1997
Buxu	5.120000	43.350000	3	GrN-19386	16,730	500	20,040	620	Solutrean	Fortea Pérez, 2001; Soto Barreiro, 2003
Cabeco de	8.937500	39.361000	CPM I lower	ICEN-542	15,820	400	19,080	450	Earland Magdalenian	Zilhao et al., 1995
Porto Marinho			CPM I lower	SMU-2015	16,340	420	19,610	520	Earland Magdalenian	Zilhao et al., 1995
Caldeirao	8.416700	39.650000	Fc	OxA-2510	18,840	200	22,623	407	Solutrean	Hedges et al., 1991; Corchón and Cardoso, 2005
El Castillo	3.965549	43.292400	8	OxA-971	16,850	220	20,230	270	Lower Magdalenian	Hedges et al., 1987; Soto Barreiro, 2003
Chufin	4.460000	43.292000	1	CSIC-258	17,420	200	20,910	240	Upper Solutrean	Straus et al., 1978; Ripoll López et al., 1999; Soto Barreiro, 2003
Cueto de la Mina	4.857000	43.426000	V (E)	Ua-3586	19,100	205	22,923	324	Upper Solutrean	Soto Barreiro, 2003; Rasilla Vives et al., 2010
Ekain	2.276000	43.236000	VII B	I-12020	16,510	270	19,790	370	Lower Magdalenian	Altuna and Merino, 1984; Soto Barreiro, 2003
			VII B	I-12224	16,030	240	19,230	310	Lower Magdalenian	Altuna and Merino, 1984; Soto Barreiro, 2003
			VII C	I-12225	15,970	240	19,170	290	Lower Magdalenian	Altuna and Merino, 1984; Soto Barreiro, 2003
			VII F	I-12566	16,250	250	19,540	580	Lower Magdalenian	Altuna and Merino, 1984; Soto Barreiro, 2003
Erralla	2.181900	43.208900	V	I 12540	15,740	240	18,940	240	Lower Magdalenian	Altuna et al., 1985; Soto Barreiro, 2003
			V	I-12551	16,400	240	19,660	340	Lower Magdalenian	Altuna et al., 1985; Soto Barreiro, 2003
			V	I-12868	16,270	240	19,500	350	Lower Magdalenian	Altuna et al., 1985; Soto Barreiro, 2003
Gato 2	1.280000	41.600000	2	GrA-22503	18,260	130	21,960	270	Solutrean	Blasco and Rodanes, 2009; Utrilla et al., 2010
	1.280000	41.600000	2	GrA-42226	17,700	70	21,115	305	Solutrean	Blasco and Rodanes, 2009; Utrilla et al., 2010
Gorham's Cave	5.342500	36.120400	III	Beta-181893	16,420	120	19,700	200	Solutrean	Finlayson et al., 2006
			III	Beta-184042	18,440	160	22,120	290	Solutrean	Finlayson et al., 2006
Hornos de la Pena	−4.029400	43.261200	B	BM-1881 R	18,450	520	22,092	652	Lower Magdalenian	Bowman et al., 1990; Soto Barreiro, 2003
La Riera	−4.856500	43.416600	17	GAK-6444	17,070	230	20,520	260	Lower Magdalenian	Straus and Clark, 1986; Soto Barreiro, 2003
			17	GAK-6445	16,900	200	20,300	240	Lower Magdalenian	Straus and Clark, 1986; Soto Barreiro, 2003
			19	GAK-6448	16,420	430	19,690	530	Lower Magdalenian	Straus and Clark, 1986; Soto Barreiro, 2003
			19	Q-2110	15,520	350	18,590	450	Lower Magdalenian	Straus and Clark, 1986; Soto Barreiro, 2003
Las Caldas	5.922200	43.336100	9	Ua-15315	17,945	370	21,541	603	Upper Solutrean	Corchón Rodrigues, 1999; Soto Barreiro, 2003
			11	Ua-15316	18,305	295	21,949	412	Upper Solutrean	Corchón Rodrigues, 1999; Soto Barreiro, 2003
			3	Land-2421	18,250	300	20,480	360	Upper Solutrean	Evin et al., 1983; Soto Barreiro, 2003
			4	Land-2422	17,050	290	21,950	380	Upper Solutrean	Evin et al., 1983; Soto Barreiro, 2003
			7	Land-2423	18,310	260	22,000	360	Upper Solutrean	Evin et al., 1983; Soto Barreiro, 2003
Mallaetes	0.300000	39.016700	III	KN-1918	16,300	1500	19,686	1723	Solutrean	Fortea Perez and Jorda Cerda, 1976
Miron	3.451900	43.245900	17	GX-25853	15,700	190	18,870	170	Lower Magdalenian	Straus and Morales, 2003
			117	GX-25857	17,050	60	20,480	80	Lower Magdalenian	Straus and Morales, 2003
			119	GX-25858	16,960	80	20,390	90	Lower Magd/Solutrean	Straus and Morales, 2003
			114	GX-28209	16,460	50	19,780	100	Lower Magdalenian	Straus and Morales, 2003
			116	GX-29439	17,400	80	20,890	120	Lower Magdalenian	Straus and Morales, 2003
			125	GX-24470	18,980	360	22,745	503	Solutrean	Straus and Gonzalez Morales, 2009
			126	GX-24471	18,950	350	22,707	514	Solutrean	Straus and Gonzalez Morales, 2009
Montlleo	1.830000	42.360000	sector B	OxA-X-2234-52	16900	110	20320	120	Solutrean	Mangado et al., 2009; Mangado et al., 2011

(continued on next page)

Table 1 (continued)

Site name	Longitude	Latitude	Level	Code	Age	Error	Cal BP	Error	Cultural attribution (sample)	References
Nerja	3.845800	36.761600	Lower 8	UBAR-158	18,420	530	22,120	630	Solutrean	Jordá Pardo et al., 1990; Jordá Pardo and Tortosa, 2008
			NV-8c	UBAR-98	17,940	200	21,590	300	Solutrean	Jordá Pardo et al., 1990; Jordá Pardo and Tortosa, 2008
			NV-8i	UBAR-157	15,990	260	19,183	315	Solutrean	Jordá Pardo et al., 1990; Jordá Pardo and Tortosa, 2008
Parpallo	0.271500	39.004100	4–4.25 m	Birm-521	17,896	340	21,488	587	Upper Solutrean	Villaverde Bonilla and Pena, 1981
			T-11	OxA-22629	18,510	100	22,115	320	Badegoulian	Aura et al., 2012
			T-16	OxA-22651	19,020	100	22,879	299	Upper Solutrean	Aura et al., 2012
Rascaño	3.700000	43.283300	4	BM-1453	15,988	193	19,150	240	Lower Magdalenian	Soto Barreiro, 2003; Aura et al., 2012
			5	BM-1455	16,433	131	19,720	210	Archaic Magdalenian	Soto Barreiro, 2003; Aura et al., 2012
Ratilla del Bubo	0.800000	38.250000	II	Land-5219	17360	180	20821	337	Upper Solutrean	Soler et al., 1990; Aura et al., 2012
Urtiaga	2.316700	43.266600	F	GrN-5817	17,050	140	20,500	160	Lower Magdalenian	Mariezkurrena, 1979, 1990; Soto Barreiro, 2003
Vale Boi	8.812320	37.090354	G25, 4	Wk-12130	18,406	164	22,080	290	Proto-Solutrean	Bicho et al., 2010, Bicho, 2004
			G25, 10	Wk-12131	17,634	108	21,160	140	Upper Solutrean	Bicho et al., 2010, Bicho, 2004

the cultural attribution of the occupation levels from which they were drawn using the Oxford Radiocarbon Laboratory database, published sources (see Table 1) and regional syntheses (e.g., Evin et al., 1983; Hedges et al., 1987; Mariezkurrena, 1990; Straus, 1990; Hedges, 1991; Hedges, 1997; Corchon Rodrigues, 1999; Soto Barreiro, 2001; Soto Barreiro, 2003; Corchón and Cardoso, 2005; Cacho et al., 2010; Aura et al., 2012; Fano, 2012). Dates were rejected if reasonable doubts concerning their validity are reported in the archaeological literature, which usually occurs when the dates are at odds with the stratigraphic position of the sample and/or the cultural attribution of associated material. The screening process reduced our original sample to $N = 30$ sites (Table 1) attributed to the Solutrean and Magdalenian cultural periods. The Lower Magdalenian, sometimes referred to as the Archaic, or Early Magdalenian, coincides with the end of the LGM on the Iberian Peninsula (Rios Garaizar et al., 2008; Aura et al., 2012; Utrilla et al., 2012) and, as a recent analysis of the temporal distribution of radiocarbon dates for the Iberian Peninsula demonstrates (Schmidt et al., 2012), Solutrean occupations in the region are coincidental with the LGM. Finally, we corrected the geo-referencing of the sites, where necessary, using satellite imagery in a GIS (ARCGIS v.10 © 1995–2012 ESRI Inc.). A statistical test (see below) indicates that the overall model of human occupation during the LGM is insensitive to the removal of any individual site.

The available sample of archaeological sites potentially under-represents Pleistocene coastal regions, parts of which were submerged after the last Glacial (Bailey et al., 2008). It may also reflect historical tendencies in the distribution of research effort and a lack of concentration on the interior (Cacho et al., 2010); an issue that we address below. Archaeological sites have a greater chance of being discovered if they are repeatedly used and archaeologists are most likely to discover sites in regions that were persistently occupied. We therefore conclude that the sample of sites available for this research reflects a more persistent pattern of occupation of the Iberian Peninsula.

Climate modeling at global and regional (Iberian Peninsula) scales

To obtain climate data for the LGM on a fine grid covering the Iberian Peninsula, the following methodology was developed:

- 1) A global low resolution (3.75° in longitude $\times$ $\sim 2.50^\circ$ in latitude) simulation, centered on 21 ka cal BP, was produced with the IPSL_CM4 AOGCM atmosphere-ocean coupled general circulation model (Marti et al., 2010). This simulation was run

according to the PMIP2 protocol (<http://pmip2.lsce.ipsl.fr/>), i.e., imposing atmospheric greenhouse concentrations of 185 ppm for CO_2 , 350 ppb for CH_4 , 200 ppb for N_2O , ice-sheets according to the ICE-5G reconstruction of Peltier (Peltier, 2004) and orbital parameters for the period 21 ka cal BP following (Berger, 1978). The simulation was run for 1000 years starting from a previous LGM simulation and is at equilibrium with its imposed LGM boundary conditions. This simulation is the official IPSL PMIP2 simulation and is further described in Kageyama et al. (2009).

- 2) The resolution of the output from the IPSL_CM4 is about 300 km on the Iberian Peninsula. To obtain high resolution data for the target region we used the downscaling method presented in (Vrac et al., 2007) to obtain monthly temperatures and precipitation on a $1/6^\circ$ grid ($\sim 16 \times 16$ km). This method involves generating rules for a General Additive Model on the basis of the IPSL_CM4 simulation of the present climate and of Climate Research Unit (CRU) data, which is on a $1/6^\circ$ grid, and then using the same rules to downscale the LGM output from the IPSL_CM4 model on the $1/6^\circ$ grid. This downscaling was first performed on an averaged annual cycle computed from 50 consecutive years of the IPSL_CM4 LGM run, following which we downscaled 50 individual consecutive years of the LGM global simulation, using the same rules, in order to be able to calculate inter-annual climate variability.

Validation of the climate model output: bioclimate testing

The downscaled climate model was initially tested by comparing the predicted average annual temperature and precipitation values for Western Europe with biome reconstructions generated from palynological samples (Wu et al., 2007). This comparison yielded satisfactory results. Next, we used archaeozoological data as proxies for climate conditions to further test the model results on the Iberian Peninsula, using a qualitative, bioclimate method as suggested by Hernandez Fernandez (2001a,b).

The mobility and environmental plasticity of large terrestrial mammals (which tend to dominate archaeozoological assemblages) makes them generally poor indicators of specific environmental conditions (Walter and Box, 1976). Animal communities, which evolve over time under the influence of climate change, are relatively conservative, however, which makes community structure a good proxy for climate conditions at a regional scale. Hernandez Fernandez (2001a) has calculated the climate tolerances (the 'index of climate restriction', or CRI) for over 1500 individual taxa of terrestrial mammal, including Chiroptera and Rodentia. The

Table 2
Bioclimate indices (BCIs) for archaeozoological assemblages dated to the LGM.

Locality	Level	I	II	II/III	III	IV	V	VI	VII	VIII	IX	Zonobiome
Abauntz	E	1.300	3.118	3.118	3.118	16.838	4.419	22.593	21.084	13.719	10.692	VI/VII
Aitzbitarte	VII	0.000	0.000	0.000	1.022	15.188	1.022	36.019	13.402	19.353	13.995	VI
Amalda	IV	0.681	1.210	1.210	2.162	18.636	1.891	33.714	9.906	19.541	11.049	VI
Amalda	V	1.970	3.322	3.322	3.465	20.100	4.813	35.170	8.826	12.187	6.826	VI
Ambrosio	IV/VI	1.430	3.970	3.970	3.970	35.726	5.399	20.728	15.728	4.540	4.540	IV
Arbreda	D	1.021	1.021	1.021	1.021	32.279	2.043	34.657	10.850	9.829	6.257	VI
Beneito	B2	0.000	0.000	0.000	0.000	38.335	0.000	25.831	25.831	5.001	5.001	IV
Ekain	VII	0.000	0.925	0.925	2.117	19.757	2.117	37.813	10.729	14.196	11.420	VI
Erralla	V	0.000	0.653	0.653	1.494	12.868	1.494	35.908	15.315	18.257	13.356	VI
Gorham's Cave	III	2.384	4.234	4.234	1.850	46.458	4.234	13.119	13.119	5.184	5.184	IV
Hornos de la Pena	C	0.000	1.587	1.587	1.587	21.112	1.587	33.019	16.352	13.965	9.205	VI
La Riera	14	0.000	0.000	0.000	0.000	20.668	0.000	37.335	20.668	10.664	10.664	VI
La Riera	15	0.000	1.586	1.586	3.630	18.391	3.630	30.294	18.391	11.246	11.246	VI
La Riera	16	0.000	0.000	0.000	2.384	19.603	2.384	33.489	19.603	11.269	11.269	VI
La Riera	19	0.000	2.776	2.776	2.776	16.108	2.776	28.614	28.614	7.779	7.779	VI/VII
La Riera	20	0.000	0.000	0.000	0.000	20.668	0.000	37.335	20.668	10.664	10.664	VI
Miron	17	0.000	0.000	0.000	0.000	28.338	0.000	33.889	13.052	17.220	7.501	VI
Miron	110	0.000	0.000	0.000	1.819	17.276	0.000	41.517	8.938	23.332	7.119	VI
Miron	111	0.000	0.000	0.000	0.000	18.891	0.000	44.816	5.001	22.591	8.701	VI
Miron	114	0.000	0.000	0.000	0.000	20.837	0.000	51.392	4.167	18.053	5.551	VI
Nerja	V-8	2.043	4.086	4.086	2.043	45.029	4.086	16.457	11.700	7.614	2.857	IV
Portalón	P1	0.000	0.000	0.000	0.000	25.005	0.000	41.668	0.000	21.664	11.662	VI
Rascano	4	0.000	3.457	3.457	3.457	15.863	3.457	34.015	15.496	13.639	7.159	VI
Rascano	5-6	0.000	1.388	1.388	1.388	15.345	1.388	38.269	17.434	15.345	8.054	VI
Urtiaga	F-G	0.000	2.287	2.287	3.240	17.348	3.240	29.902	15.461	16.728	9.507	VI
Vale Boi	Solut.	1.300	5.427	5.427	5.427	32.782	6.727	17.327	14.300	7.155	4.127	IV

The BCI scores are calculated for each of Walter's zonobiomes (roman numerals). The dominant zonobiome, indicated by the highest BCI score, is indicated in the final column. The provenience of each archaeozoological assemblage is indicated by site and by level.

CRI scores reflect the probability of encountering a taxon in each of the nine global ecological climate-zones, or zonobiomes (Walter and Breckle, 1985; Walter and Breckle, 2000; Hernandez Fernandez, 2001b). Cumulative probability scores for each zonobiome are calculated on the basis of the presence/absence of taxa in a faunal assemblage and associated CRI scores. The resulting score, the Bioclimate Index (BCI), represents the probability of occurrence for each of the zonobiomes. The highest BCI score characterises the environment within which the original biocenosis existed; where two BCI scores are almost equal, a zonoecotone is defined. This method has the advantage of considering micro- and macro-mammals, thus controlling for different spatial scales of environmental sensitivity as well as making use of the presence/absence data common in the archaeological literature.

We calculated cumulative BCI scores for archaeological levels dated to the LGM for which faunal remains are published (Table 2). The zonobiome with the highest score was retained for testing (Table 2). The zonobiomes with the highest probability of being present in our LGM archaeological sample on the basis of the BCI calculations are: IV (Arido-Humid Winter Rain Region), and VI (Temperate Nemoral Climate) and, in two cases, 'zonoecotone' VI-VII (Forest-Steppe) (Fig. 2). Other zonobiomes indicated secondarily include: VII (Arid-Temperature Climate) and VIII (Cold-Temperature Boreal Climate). The distribution of zonobiomes suggested by our results is compatible with Brugal and Yravedra's biogeographical zonation of Southwest Europe (Brugal and Yravedra, 2005/2006). Next, we estimated the temperature and precipitation intervals for the zonobiomes, as described below.

Zonobiomes are characterised by the presence of zonal vegetation types and the climate conditions that gave rise to them (primarily temperature and precipitation). Since the distribution of the world's vegetation zones is continuous, the zonobiomes grade into one another; where they intersect zonoecotones are formed, i.e., transitional regions where the vegetation from two distinct zonobiomes either mixes or forms a macro-mosaic, usually as a result of a micro-climate induced by relief and/or soil type (Walter and Box, 1976; Walter and Breckle, 2000). Climate conditions for the

zonobiomes are traditionally defined using climatograms (standardized ecological climate diagrams) for type locations. Defined climate intervals are needed to test the climate simulation results, however. We generated these intervals using data from ground climate stations selected using the current distribution of the zonobiomes. Since climate conditions within a globally distributed zonobiome vary with location (particularly with hemisphere and latitude), we used climate stations in the Northern Hemisphere within or near our target region. Climate readings for 30-year intervals were downloaded from the Climate Research Unit's on-line database⁴ and the calculated intervals were used to estimate climate ranges for the LGM at our sample locations, based on the BCI results (above) and then used to test the goodness-of-fit of the downscaled climate simulation. These tests were conclusive.

Quantifying climate variability during the LGM over Iberia

We used the 50-year series of monthly averages of temperature and precipitation obtained by downscaling the global model results on the Iberian Peninsula (above) to calculate climate variability indices. The variability indices for temperature (temp_var) and precipitation (precip_var) are calculated on the high resolution grid covering the Iberian Peninsula as follows: 1) annual averages are calculated for each point, resulting in a three-dimensional matrix of 50 values on a 16 × 16 km grid; 2) a regional mean and percentile value are calculated from this matrix; 3) each point in the matrix is then re-classified as either 'average' (score = 0), 'severe' (1 standard deviation from the mean, score = 1) or 'extreme' (2 standard deviations from the mean, score = 2). The resulting 16 × 16 km grid of single values is then interpolated using the Natural Neighbour tool in ARCGIS v.10 (at a spatial scale of 10 km) to produce the variability surfaces (Fig. 3a, b) for easy comparison to other data used in our analyses: e.g., topography, hydrography and archaeological sites.

⁴ Source: <http://www.cru.uea.ac.uk/>.

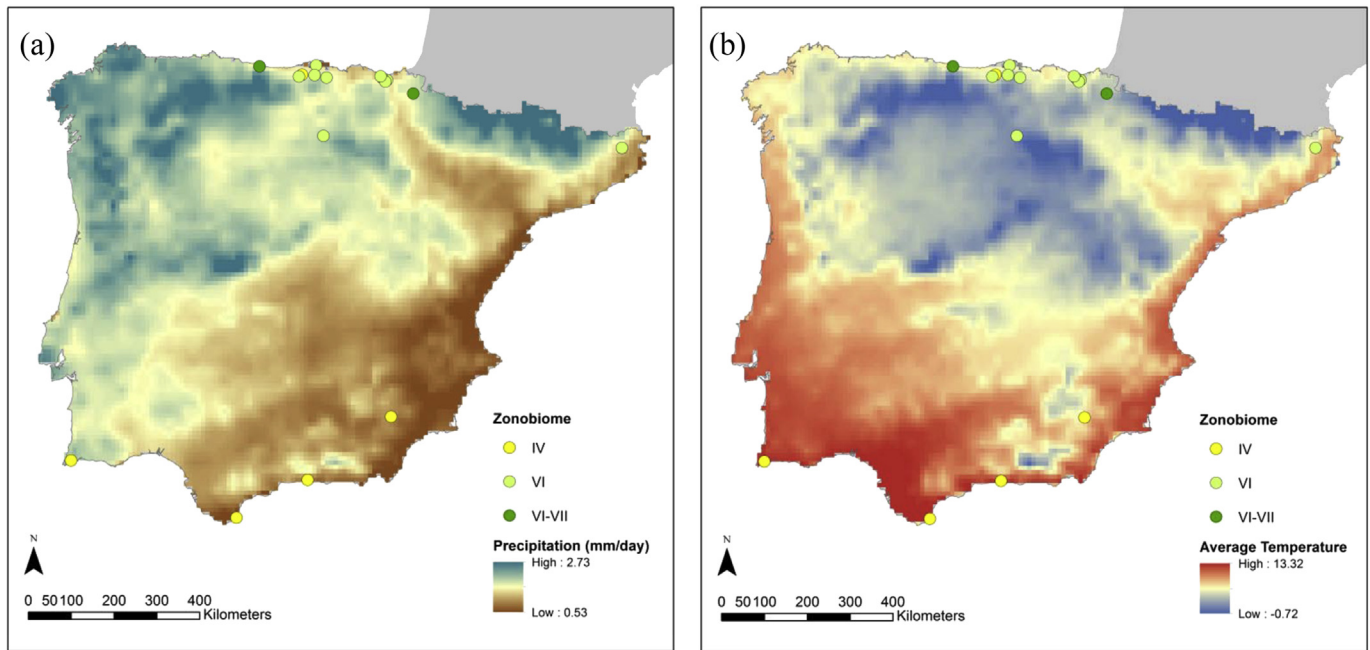


Figure 2. Dominant zonobiomes indicated by BCI values calculated for archaeozoological assemblages dated to the LGM against: average annual precipitation in mm/day (a) and average temperature in °C (b) for the LGM. Palaeoclimate averages are calculated for a 50-year period.

Next, we calculated the Standard Precipitation Index (SPI) using a 12-month interval. Originally developed as a tool for water management in Colorado (U.S.A.), the SPI reflects the impact of variable precipitation rates on water supply and has proven useful as a means of quantifying drought events (McKee et al., 1993; Guttman, 1999). The SPI for a given monthly precipitation value is calculated as a function of the cumulative probability distribution of previous monthly precipitation rates for a specified timeframe, transformed to a normal distribution. The transformation is necessary since precipitation rates are rarely normally distributed at shorter timeframes (i.e., three, six and twelve months). Since the SPI is normally distributed it can be used to quantify both wet and dry events and it has the advantage of only requiring a single predictor.

The SPI source code was downloaded from the Colorado State Climate Center web-site.⁵ The SPI indices were calculated month by month for each point on the grid, for the 50-year time series, and each month was reclassified according to the Global Standardized Precipitation Index.⁶ Four different raster maps were produced using the reclassified values, summing the number of months classified as 'extreme', i.e., SPI indices below -2.0 ('extremely dry') or above 2.0 ('extremely wet'); or classified as 'severe', i.e., indices below -1.5 or above 1.5 (combining severe and extreme values). Each of the resulting grids was then interpolated, as above. The resulting predictor variables, based on the SPI, are: 'SPLevt' (a tally of the number of months for which extreme conditions, either wet or dry, existed), and 'SPLev' (a tally of the number of months for which SPI values are less than -1.5 or greater than 1.5). We also calculated separate predictors for severe and extreme wet conditions, and for severe and extreme dry conditions using the SPI. 'SPLev' reflects the number of months in a 50-year period during which unusually severe to extremely dry conditions existed, i.e., the frequency of drought events (see Fig. 4).

Statistical analysis

Next, we constructed and compared several logistic regression models to determine which environmental and climatic variables best described the presence of archaeological sites dated to the LGM. For predictor variables, we used a 50-year series of high-resolution climate data (16×16 km, monthly time-steps) over the Iberian Peninsula, obtained by downscaling the results of a global atmosphere-ocean general circulation model run at a ~ 300 km resolution, to quantify inter-annual climate variability during the LGM for precipitation and temperature. We also calculated precipitation extremes using McKee's Standard Precipitation Index (SPI) (McKee et al., 1993) and quantified inter-annual variability for this index, as described above. Elevation data were derived from the SRTM 90-m digital elevation model⁷; the hydrological data were derived from the HydroSHEDS dataset.⁸ The geographical variables used in the spatial analysis include: elevation (elev), aspect (divided into eight classes), slope and distance to nearest water (using a cost-surface generated in ARCGIS 10 and the Hydro-1 Kilometer Watershed Model); the climate variables used include: mean annual temperature (temp) and precipitation (precip) for the LGM, cumulative annual precipitation, temperature and precipitation variability (temp_var and precip_var) on an inter-annual time-scale and the SPI predictors described above.

Georeferenced archaeological sites ($N = 30$ sites) securely dated to the LGM (Table 1) were used as the presence component of the binary response variable for logistic regression. Two different approaches were used to include absences for these models: 1) random pseudo-absences ($N = 1000$) and; 2) 50 archaeological sites attributed to the Middle Palaeolithic (MP) and not occupied during the LGM. Pseudo-absences were randomly selected from all possible locations within the study region restricted to locations farther than 15 km (i.e., beyond nearest neighbour on a 10 km grid).

⁵ Source: <http://ccc.atmos.colostate.edu/>.

⁶ Source: <http://iridl.ldeo.columbia.edu/maproom/Global/Precipitation/SPI.html>.

⁷ Source: <http://www2.jpl.nasa.gov/srtm/dataproduct.htm> (Jet Propulsion Laboratory, CalTech).

⁸ Source: <http://gisdata.usgs.gov/website/HydroSHEDS/> (U.S. geological survey).

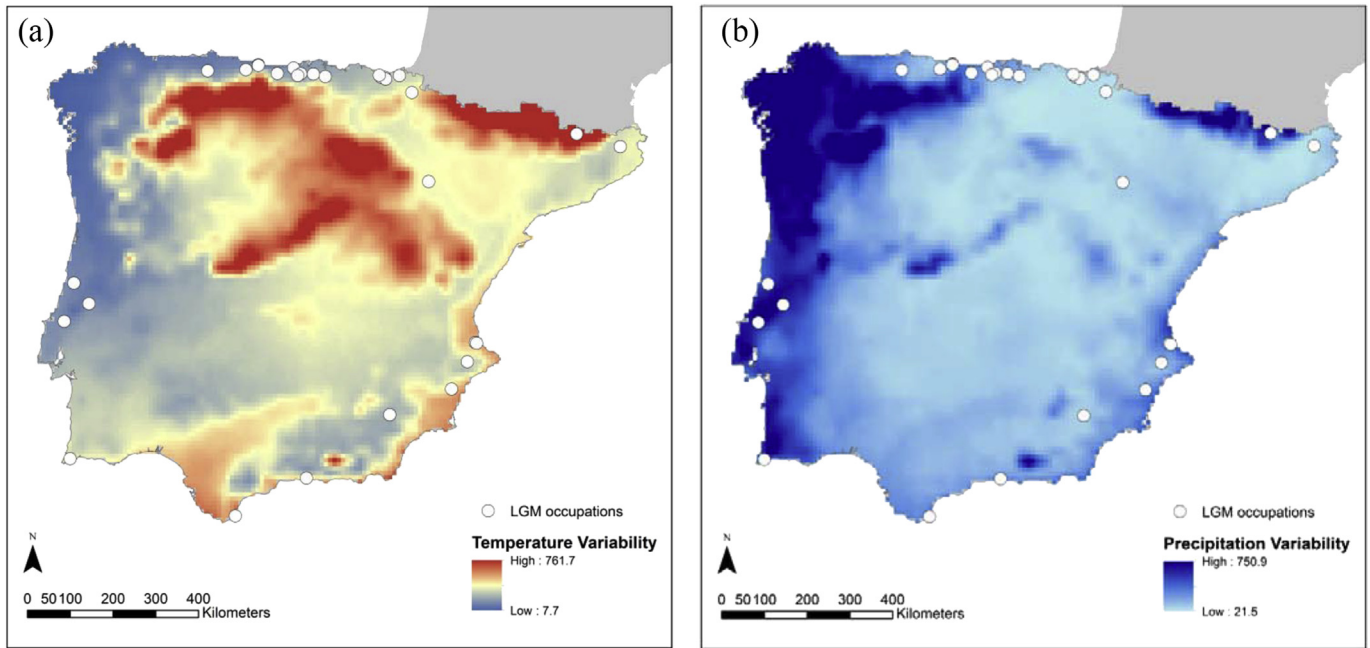


Figure 3. Variability surfaces for temperature (a) and precipitation (b) during the LGM. 3a: warm colours indicate more frequent anomalies in monthly temperature compared with the 50-year average for each 10×10 km cell, cool colours less frequent anomalies; 3b: dark tones indicate progressively more frequent precipitation anomalies over the same period. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).

from known LGM archaeological sites (presences). We tested 100 (replications = 100) different random subsets of 1000 locations that represented absences in our series of regressions to identify the best model. The 50 MP sites used as 'true' absences were randomly selected from the PACEA database (d'Errico et al., 2011). This second test-set contains 'true' null values for the LGM because it consists of

locations investigated archaeologically and therefore known to be unoccupied during the LGM, as opposed to our sets of pseudo-absences (which are not known to be occupied during the LGM but may or may not have been investigated). The second test-set (MP sites) has a restricted geographical distribution similar to that of the LGM dataset (Fig. 1) and thus removes the potential bias that may have been introduced into the analysis by the uneven distribution of archaeological research effort (historically, surveys have tended to concentrate in areas previously identified as rich in archaeological sites).

We tested the relationship between the presence of archaeological sites at the LGM and multiple environmental predictors using logistic regression and both approaches to modeling absences. Multiple models were compared that included different combinations of geographic and environmental predictors, representing plausible hypotheses regarding the influence of spatial and climatic variability on the spatial distribution of human populations. The best model was selected from amongst the candidate models using the sample-size corrected Akaike Information Criterion (AICc) (Burnham and Anderson, 2002). The models tested combine geographical and environmental predictors of site location tested independently and in combination. A selection of models and the complete list of predictors tested are presented in Table 3A. In testing multiple sets of random pseudo-absences, we created a distribution of $\Delta AICc$ values (mean and standard deviation). The model that had the lowest $\Delta AICc$ over all 100 iterations of pseudo-absences was selected as the 'best' model.

Once the best model was identified using AICc in the first analysis, model performance was assessed using the receiver operator curve (ROC) and the associated metric, area under the curve (AUC). The ROC analysis is used in predictive modeling to assess model performance and sensitivity (Fielding and Bell, 1997; Guisan and Zimmermann, 2000; Boyce et al., 2002; Cullingham et al., 2012; Jiménez-Valverde, 2012). The ROC curves are produced by plotting the false-positive rate against the true positive rate at

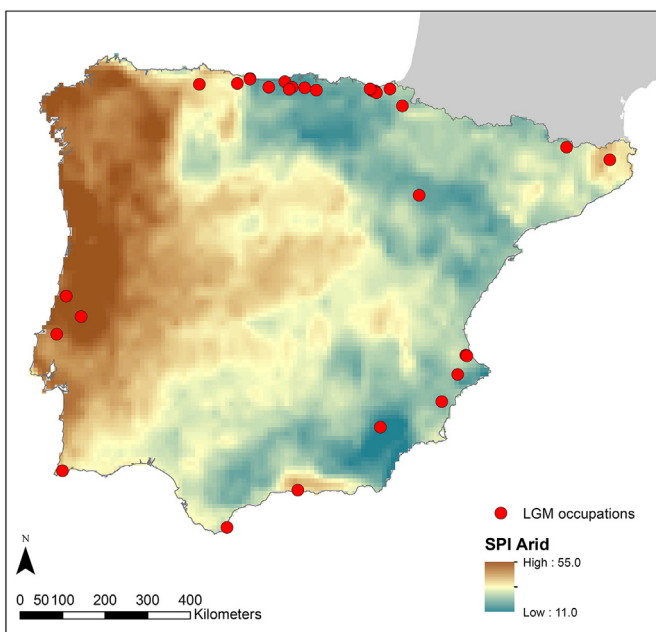


Figure 4. Spatial distribution and frequency of extremely dry SPI values ('SPIarid'). Warm colours indicate a higher frequency of occurrence of precipitation anomalies leading to drought events, cool colours indicate a lower frequency. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).

Table 3A

Summary of models tested based on LGM sites (presences) versus random pseudo-absences.

Model ID	Model	K*	Mean Δ AICc	Mean model likelihood	Mean AICc weight
Mod.12	SPLarid + elev + slope	4	0.001 (0.007)	1 (0.003)	0.676 (0.122)
Mod.9	temp + elev + slope	4	4.208 (1.983)	0.189 (0.11)	0.11 (0.085)
Mod.11	SPIsev + elev + slope	4	4.279 (1.115)	0.138 (0.092)	0.088 (0.048)
Mod.10	precip_var + elev + slope	4	4.669 (1.162)	0.113 (0.061)	0.071 (0.029)
Mod.7	elev + slope	3	5.284 (1.543)	0.092 (0.066)	0.056 (0.029)
Mod.8	SPLarid + slope	3	52.346 (3.192)	0	0
Mod.5	Slope	2	56.286 (3.304)	0	0
Mod.2	elev	2	61.61 (2.705)	0	0
Mod.4	SPLarid	2	83.028 (3.051)	0	0
Mod.3	SPIsev	2	92.02 (2.956)	0	0
Mod.6	temp	2	93.455 (2.869)	0	0
Mod.1	precip_var	2	94.39 (2.908)	0	0

Model comparisons were repeated 100 times using different random subsets of points as absences ($N = 1000$). Numbers in parentheses represent standard errors of the parameter estimates over 100 replicates. K indicates the number of parameters in the model. Mean Δ AICc refers to the average difference between each model's AICc and the minimum AICc value among all models tested over 100 replicates using different sets of pseudo-absences. A summary of the best model (mod.12) is presented in Table 3B.

different thresholds. The area under the ROC curve (the AUC) is a measure of overall model performance, i.e., the probability that the model is making correct predictions. A series of models was built using 60% of the full data set and performance was assessed by comparing the resulting predictions for the remaining 40% to their actual observed values. Bootstrapped comparisons were repeated 999 times to generate a distribution of AUC values using different training data sets. Finally, we tested the individual impact of each archaeological site on the best model, calculating the difference between the AUC of the model and the AUC of the model minus the site (the marginal impact). Logistic regression was performed using the glm function in R, and ROC analysis was undertaken using the ROCR package (Sing et al., 2009).

Results

The best model of site location during the LGM, based on a comparison of LGM sites against a set of randomly selected points on the landscape, combines the geographical predictors slope and elevation with the climate predictor 'SPLarid', which measures the frequency of occurrence of precipitation anomalies leading to unusually dry conditions in a given location during a 50-year period using the SPI (SPI values < -1.5) (Table 3A,B; model 12; Fig. 5). This model is the most parsimonious when comparing LGM sites to the background environment (the population of random "null" values), based on AICc (Akaike, 1974). All predictors in this model were

Table 3B

Summary of logistic regression parameter estimates for the best model indicated in Table 3A (Mod.12).

	Estimate	Standard error	Z	Pr ($> z $)
(Intercept)	−0.915 (0.027)	0.894 (0.003)	−1.029 (0.032)	0.327 (0.016)
SPLarid	−0.065 (0.001)	0.026 (0)	−2.493 (0.024)	0.015 (0.001)
Elev	−0.005 (0)	0.001 (0)	−5.061 (0.013)	<0.001
Slope	0.244 (0.001)	0.043 (0)	5.629 (0.01)	<0.001

Numbers in parentheses represent standard errors of parameter estimates based on 100 replicates of the model using different random subsets of points as pseudo-absences.

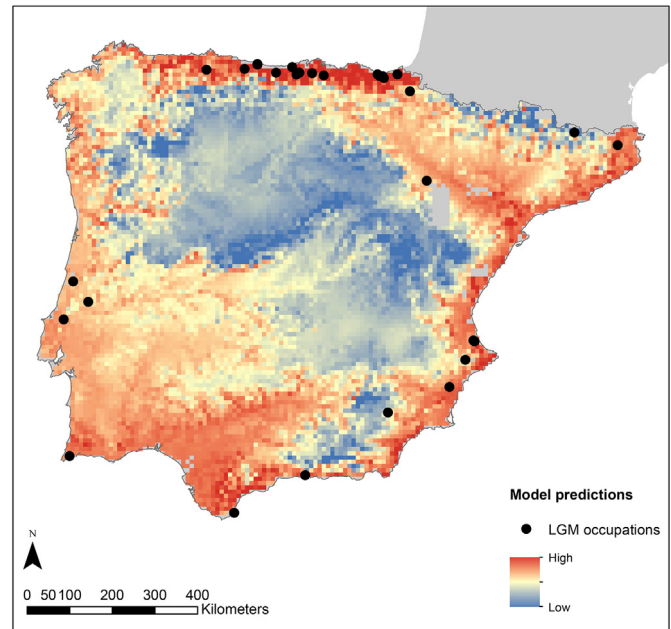


Figure 5. Predicted probability of occurrence of archaeological sites based on model 12 (Table 3B). The predicted probabilities have been log-transformed. Warm colours indicate a high probability and cool colours indicate a low probability of occurrence based on this model. Black points represent known LGM presences which were used with replicated sets ($n = 100$) of random pseudo-absences to build a predictive logistic regression model. This predictive surface uses the average coefficient estimates from the replicate models. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).

found to be statistically significant (Table 3A). Validation of the model was done using ROC and the resulting mean AUC of 0.87 ($SE = 0.0046$) indicates that model predictions are accurate 87% of the time. The marginal impact values obtained for each archaeological site indicate that removal of any individual site has an insignificant impact on the model: δ AUC values ranged between -0.001 and 0 (mean = -0.0006 , $SE = 0.00001$).

Climate averages (average annual precipitation, average annual temperature) are not significant predictors, leading us to conclude that climate variability, rather than average climate conditions, is affecting the spatial distribution of human populations at the scale of the Iberian Peninsula during the LGM. The model described above suggests that at the scale of the Iberian Peninsula, humans were preferentially selecting regions less frequently affected by drought.

Table 4A

Summary of logistic regression models tested based on LGM sites as presences and MP sites as absences ($n = 50$). K indicates the number of parameters in the model. Δ AICc refers to the average difference between each model's AICc and the minimum AICc value among all models tested. A summary of the best model (mod.3) is presented in Table 4B.

Model ID	Model	K*	Δ AICc	Model likelihood	AICc weight
Mod.3	SPIsev	2	0.000	1.000	0.460
Mod.11	SPIsev + elev + slope	4	1.154	0.562	0.259
Mod.4	SPLarid	2	4.037	0.133	0.061
Mod.8	SPLarid + slope	3	4.681	0.096	0.044
Mod.5	Slope	2	4.806	0.090	0.042
Mod.1	Precip_var	2	5.520	0.063	0.029
Mod.12	SPLarid + elev + slope	4	5.585	0.061	0.028
Mod.7	Elev + slope	3	6.092	0.048	0.022
Mod.10	Precip_var + elev + slope	4	6.109	0.047	0.022
Mod.2	Elev	2	7.168	0.028	0.013
Mod.6	Temp	2	7.435	0.024	0.011
Mod.9	Temp + elev + slope	4	7.851	0.020	0.009

Table 4B

Summary of logistic regression parameter estimates for the best model indicated in Table 4 (Mod.3) using MP sites as absences. See text for details.

	Estimate	Standard error	z	Pr ($> z $)
(Intercept)	4.261	1.771	2.407	0.016
SPIsev	−0.063	0.024	−2.646	0.008

When we use the second approach to modeling absences (MP sites unoccupied during the LGM) in the logistic regression, we obtain fairly consistent results (Table 4A, B). Because the distribution of MP sites is very similar to that of LGM sites (Fig. 1), this analysis is more spatially restricted than the previous one, i.e., it takes place at a more regional scale. The best model in this second statistical analysis is a single predictor, ‘SPIsev’, which reflects the frequency with which severe precipitation anomalies occur at both ends of the scale (wet and dry) (Fig. 6). Elevation (elev) and slope are no longer significant predictors in this analysis, which could indicate that locations were selected on the basis of similar geographical variables before and during the LGM. This model performed less well than our model using pseudo-absences and had an AUC of 0.71. The results of the two statistical analyses are consistent and indicate that inter-annual climate variability (specifically in precipitation) is a potentially significant factor affecting the distribution of human populations in Iberia during the LGM.

Discussion

It is possible that the concentration of research effort in regions historically considered archaeologically interesting has created an inherent bias in the archaeological record (Cacho et al., 2010; Canals et al., 2010; Cacho et al., 2012). Using two approaches to modeling absences (randomly generated pseudo-absences versus MP sites unoccupied during the LGM) allows us to test for potential biases in the archaeological record and therefore, in the statistical analysis presented above. The chances of discovery for LGM and MP sites

will have been similarly affected by the regional history of archaeological investigation, which is why their spatial distribution is very similar (Fig. 1). A comparison of the results of the statistical analysis of the distribution of LGM sites against both sets of absences, therefore, should reveal the presence of bias in the archaeological record. The results using both approaches are consistent and indicate that inter-annual climate variability is a significant predictor of the presence of sites during the LGM. This suggests that biases inherent in the archaeological record are not hindering our analysis.

Since the results of the analysis using pseudo-absences are more robust (as indicated by the validation procedure), we consider the model resulting from this analysis to be the best one to adopt. We propose, therefore, a statistical model of archaeological site location for the LGM that combines geographic variables (elevation + slope) with a climate variable (‘SPIlarid’) that describes the frequency with which drought events occur (Fig. 5).

We test annual temperature and rainfall averages during the LGM, as well as cumulative precipitation values, and show that they do not appear to have affected the distribution of modern human populations in Iberia significantly. These results are consistent with previous spatial analyses conducted at lower spatial and temporal resolution (Davies and Gollop, 2003; Banks et al., 2009). Davies and Gollop (2003), working with climate simulation ‘snapshots’ intended to capture millennial-scale variation at a spatial scale of 60 km, found that average temperature during the LGM was not a good predictor of site location in Western Europe during the last Glacial; our results agree with theirs. Banks et al. (2009) model the ecocultural niche of Middle and Late Solutrean populations during Heinrich Event 2 (HE2) and the LGM using archaeological assemblages as proxies for human populations. The authors incorporate millennial-scale variability for a suite of climate variables (i.e., average annual temperature, coldest/warmest months, average annual precipitation) at a spatial resolution of 60 km into their statistical model. The resulting ecocultural niche reconstructions (Banks et al., 2009: Fig. 2) suggest that the Iberian Peninsula provided favourable conditions for Solutrean populations during the LGM. Our results are broadly compatible with this hypothesis, while highlighting the existence of spatial heterogeneity at a finer resolution (both temporal and spatial), which appears to have had an impact on the distribution of human populations within the available niche space.

Our model is consistent with the hypothesis that the Iberian Peninsula formed a refugium for hominins (and other taxa) during maximum glacial conditions (Straus et al., 2000; Gamble et al., 2004; O’Regan, 2008; Verpoorte, 2009). Other authors have also observed that the Iberian Peninsula was a spatially heterogeneous landscape during the LGM (O’Regan, 2008). This spatial heterogeneity could explain why northern Spain, in particular, is believed to have acted as an important population reservoir for human populations during the LGM (Straus, 2000; Gamble et al., 2004), since it was a favourable ecological zone with relatively low levels of climate variability (more precisely, a low frequency of drought events). At the same time, our model indicates that favourable conditions also existed elsewhere on the Iberian Peninsula, e.g., in southern Iberia (including the Guadalquivir Basin) and along the Mediterranean coast.

Our results are also not incompatible with the suggestion that southern Iberia had a relatively ‘buffered’ climate (Jimenez-Espejo et al., 2007), since climate variability is less pronounced in this region. This observation can also be extended to other parts of Iberia on the basis of our model (Fig. 5; and see Figs. 3, 4 and 6), however, the case for southern Iberia as a special climatic zone (Jimenez-Espejo et al., 2007), at least during the LGM, is therefore somewhat weakened.

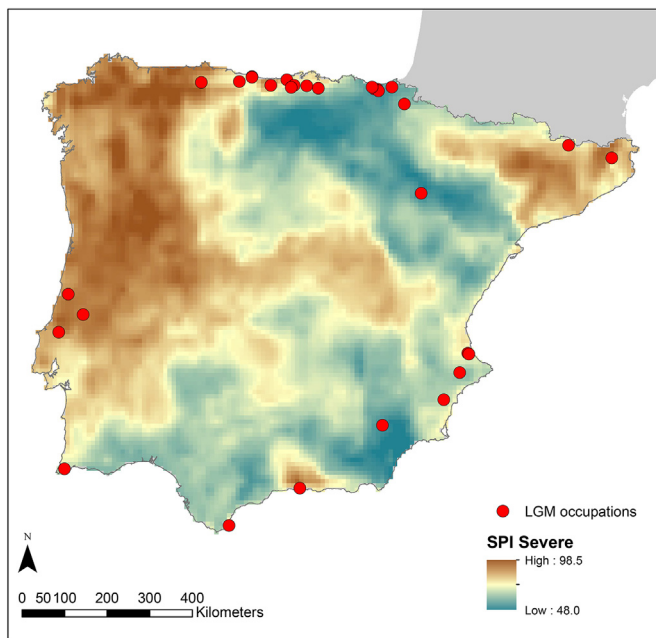


Figure 6. Spatial distribution and frequency of severe SPI values (both wet and dry events). Warm colours indicate a higher frequency of occurrence of anomalies, cool colours indicate a lower frequency of occurrence. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Bicho and colleagues argue that the coastal zone of southwest Iberia was a highly productive environment due to coastal upwelling, particularly during cold phases of the last Glacial (Bicho and Haws, 2008; Haws et al., 2011). The archaeological record provides evidence for human exploitation of this coastal zone during the LGM (e.g., the site of Vale Boi) although the evidence is not overwhelming. This could be due to the inundation of coastal regions at the end of the Pleistocene, however (Bailey et al., 2008; Bicho and Haws, 2008). Our model suggests that, in addition to including a highly productive coastal zone, southwest Iberia was a highly suitable environment because of its relatively equable climate.

We are not suggesting that regions with less favourable conditions (according to our model) were completely unoccupied. The archaeological record contains evidence of human occupation attributable to the LGM on topological grounds in several of these regions [e.g., the Central Meseta (Cacho et al., 2010, 2012) and Galicia (Valcarce, 2011)]. We agree with previous observations (Straus, 2000; Straus et al., 2000) that these sites, which are frequently open-air occupations of relatively short duration, probably indicate the ephemeral, potentially seasonal occupation of relatively less favourable regions by populations more permanently based elsewhere.

Finally, our results are consistent with the suggestion that dry conditions during cold phases of the last Glacial played a role in regulating population distribution in Iberia (Jimenez-Espejo et al., 2007; Sepulchre et al., 2007; Bradtmöller et al., 2012). These authors define dry conditions in terms of millennial scale fluctuations in global precipitation levels at low spatial resolution, whereas we test average annual precipitation, cumulative precipitation, inter-annual variability in precipitation levels and variability in SPI at a fine spatial resolution (10×10 km). Average annual precipitation levels, cumulative precipitation rates and inter-annual variability in precipitation rates measured against a regional average (precip_{var}) did not prove to be significant predictors of archaeological site location. Using the SPI to examine precipitation rates, however, reveals that the frequency of drought events ('SPIdrid') and to a lesser degree, the frequency of extreme precipitation events, both arid and wet ('SPIsev'), appears to have an impact on the distribution of archaeological sites. This suggests that drought events are the yardstick against which 'dry' conditions should be measured.

Droughts are geographically restricted, local events of variable duration (Dai, 2011), creating a local scarcity in water resources essential to maintaining human life. Droughts affect productivity through their effect on vegetation growth and biomass (Chen et al., 2012). As a result, droughts also affect the distribution of ungulate populations (Duncan et al., 2012), a key resource for prehistoric populations living in Iberia during the LGM. The likelihood of drought events occurring, therefore, creates uncertainty that can be translated in human terms as an inability to accurately predict the resources that will be available after a residential move - a potentially fatal situation given the attendant costs of mobility for hunter-gatherers. Our results suggest, therefore, that regions relatively more affected by climate variability, particularly in the form of frequent drought events, may have failed to support a persistent human population during the LGM due to unacceptably high levels of ecological risk.

Our model shows that although the Iberian Peninsula may have been a relatively hospitable environment during the LGM, in keeping with its probable role as a Glacial refugium, it was a spatially heterogeneous landscape in terms of climate variability. This heterogeneity, in turn, likely created spatial discontinuities in the distribution of human populations that are visible in the archaeological record. Discontinuities in the pattern of human settlement would have created gaps in both gene flow and cultural

transmission within the Iberian Peninsula, which could explain the existence of regional, typological differences between Solutrean assemblages (Straus, 2000) [although an alternative explanation for these regional differences has been offered (Banks et al., 2009)].

Conclusion

In conclusion, this research introduces a novel methodology for the creation of high-resolution palaeoclimate models, testing the resulting climate simulations using archaeological data and quantifying small-scale (inter-annual) climate variability. Thus, we are able to statistically test the impact of a range of environmental variables on the spatial distribution of prehistoric populations at a much finer scale, both temporally and spatially, than previously possible. This has enabled us to demonstrate that inter-annual climate variability at a fine spatial scale (10×10 km) affected the spatial behaviour of prehistoric human populations during the LGM in Iberia.

The ramifications of this research in terms of the Variability Selection Hypothesis (Potts, 1996, 1998, 2013) include a demonstration that hominins (in this case modern humans) are sensitive to environmental variability, as predicted. At a fine scale of spatial and temporal resolution, however, and under specific conditions (the LGM in the Iberian Peninsula), humans appear to avoid regions where variability is relatively high, which runs counter to predictions. This indicates a need for further research into the impact of environmental variability at different resolutions, and on different groups of hominins, in order to establish the spatial and temporal scale at which Variability Selection operates. Studies similar to the one presented here need to be conducted, testing the limits of our lineage's tolerance for climate variability. This will also help us better understand the pattern of hominin dispersal, which is considered within Variability Selection. The next step in our research, therefore, will be to test the impact of climate variability during MIS 3 on the pattern of human dispersals into and within the Iberian Peninsula using the methodology established in this study and new high-resolution numerical climate experiments.

Acknowledgements

This research was made possible by a grant from the Fonds de recherche du Québec - Société et culture (FQRSC) #SE130925, which supports the activities of the Hominin Dispersal Research Cluster, and additional support from the Social Sciences and Humanities Research Council of Canada #401-2009-2240. The LSCE team thanks the Centre de Calcul Recherche et Technologie for providing the computing time required to perform the global LGM simulation presented in this work. In addition, the authors would like to thank the anonymous reviewers for their thoughtful reviews.

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